

# Vignesh Jagdish Battulkar

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Working eligibility: Stamp 1G Visa

www.linkedin.com/in/ vignesh-battulkar-548a1b1b8 Vanity URL name

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## Professional Summary

Data Analytics graduate with a Master's degree in Artificial Intelligence, experienced in data analysis, dashboarding, data quality improvement, and insight generation across large datasets. Strong background in Python, analytics, and visualisation tools, with hands-on experience supporting business performance reporting and stakeholder decision-making. Demonstrated ability to conduct deep-dive analyses, improve metric integrity through structured preprocessing, and collaborate effectively in cross-functional environments. Motivated to contribute to Mastercard Launch by driving data-driven insights and continuous improvement initiatives, including the use of AI to enhance product performance and customer experience.

## Education

**Dublin City University**, Dublin, Ireland

Sept 2024 – Sept 2025

**MSC in Computing Major Artificial Intelligence** - Relevant Coursework: Machine Learning, AI, Data Mining and Data Analytics, AI Information Seeking, Data Management, Statistical data analysis, Professional Research, Computer Vision.

**Result** - 2.1 Honours

**KLE Society's RLSIBCA College**

Aug 2020 – Aug 2023

**Bachelor's degree in Computer Application (Distinction)** - Relevant Coursework: Data Structures, C++, Java, Python, Cloud Computing, Data Mining, Software Development Engineer, RDBMS, SDLC life cycle, OOP, Algorithms, Time/Space Complexity

**Result** - 78% Distinction

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## Technical Skills

**Data and Analytics:** Data Analysis, Machine Learning, Data Quality Assessment, Feature Engineering, and Data visualization

**Soft skills :** Teamwork and collaboration, Communication and curiosity, Adaptability and Flexibility, Strong Ownership and Continuous Learning Mindset, Time Management and problem-solving, Analytical Thinking, Decision-Making

**Languages and Databases :** Python, java(basics), SQL

**Platforms :** AWS, Microsoft Azure, Oracle

**Tools:** Power BI, Tableau PyCharm, MS Excel, MS Office.

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## Experience

### Deloitte Data Analytics Virtual Internship

- Managed, cleaned, and analyzed employee compensation data to assess pay equality across factories and job roles. Applied advanced Excel and reporting skills to deliver business insights. Demonstrated initiative by independently managing tasks while contributing to overall project goals.
- **Tools:** Excel, Python, Tableau.

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## Projects

### Hate Speech Detection (Text + User Interactivity)

- Five models (Random Forest, Naive Bayes, SVM, BiLSTM, BERT) were compared and developed to automate hate speech detection on over 200,000 tweets, using different engagement metrics (likes, retweets, timestamps).
- Performed extensive data cleaning, feature engineering, and quality checks to ensure metric integrity. Evaluated multiple ML models and compared results to support data-driven decision-making.
- Random Forest achieved 96% accuracy (the most interpretable models), BiLSTM achieved 94 (best for contextual modeling) and BERT had an accuracy of 88.7% (deep understanding of semantics).

- Improved data quality and metric integrity through structured data cleaning, validation, and feature standardisation prior to analysis and reporting.
- These project outcomes provided nearly full impact (approx. 96%) and implications for data driven approaches to decision making for content moderation and the ethical expectations of content moderation were introduced postexperiment to considerations and applications (as well as scalability).

#### **CBI-Aligned Banking Compliance Assistant(AI + RAG + ML + Streamlit Deployment Project)**

- Built an end-to-end Retrieval-Augmented Generation (RAG) system using OpenAI embeddings and FAISS to enable semantic search over Central Bank of Ireland and EU regulatory documents.
- Implemented a machine learning-based query classification model to route compliance questions into AML, GDPR, Consumer Protection, and Governance domains for more accurate retrieval.
- Developed and deployed an interactive Streamlit web application with citation-grounded LLM responses, ensuring answers are strictly derived from regulatory sources.
- Applied practical knowledge of NLP concepts including embeddings, vector search, chunking strategies, and similarity scoring.
- Demonstrated structured problem-solving and independent learning by designing, building, and deploying a productionstyle AI compliance assistant.

#### **Amazon Review Rating Prediction (ML & Deep Learning)**

- Worked alongside a team of 4, tight deadlines, complex ideas and effective team and individual work.
- Conducted deep-dive analyses on large datasets to uncover actionable insights that informed data-driven decisions.
- Communicated results in clear reports and presentations, by demonstrating excellent verbal and written skills.
- Utilized NLP (BiLSTM with GloVe embeddings) and Achieved 77.3% classification accuracy and troubleshoot performance issues, ensuring reproducibility and clear documentation.
- **Technologies & Tools:** Python, TensorFlow, GloVe, scikit-learn, pandas, NumPy, Matplotlib

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### **Certification**

#### **AWS- Getting started with Storage, cloud technology**

- Gained hands-on experience with Amazon S3 buckets and cloud storage services, applying cloud concepts to practical data management tasks.

#### **Microsoft Azure – Fundamentals of Machine Learning**

- Learned core ML concepts, model training workflows, and deployment fundamentals using Azure ML Studio.

#### **Google Cloud – Applying AI Principles**

- Explored AI solution design on Google Cloud, focusing on ethical AI principles, scalability, and integration with business processes.

#### **Deloitte – Data Analytics Job Simulation**

- Completed a simulation analysing HR data, developed fairness classification logic, and built Tableau dashboards for business insights.